

Ref. No. CECNF/N/2024-25/21018

U.G. Semester-II Examination, 2024-25 (NEP)

STATISTICS

Paper: STAMJ21 / STAMN21 — Introduction to Probability Theory

Time: 2:30 Hours

Full Marks: 70

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*Note: Attempt any **five** questions. Each question carries equal marks.*

**Question 1.**

- (a) A fair die is tossed three times. Let  $A_i$  be the event that  $i$ -th toss turns up with even number,  $i = 1, 2, 3$ . Write down the events in terms of  $A_1, A_2$  and  $A_3$  that:

- (i) at least one of the tosses turns up with even number.
- (ii) exactly one of the tosses turns up with even number.
- (iii) not more than two of the tosses will turn up with even number.

- (b) If  $S = \{x : 0 < x < 12\}$ ,  $M = \{x : 1 < x < 9\}$  and  $N = \{x : 0 < x < 5\}$ , then find:

- (i)  $M \cup N$
- (ii)  $M \cap N$
- (iii)  $M^c \cap N^c$

- (c) A lot consists of 10 good articles, 4 with minor defects, and 2 with major defects. One article is chosen at random. Find the probability that:

- (i) it has no defects,
- (ii) it has no major defects,
- (iii) it is either good or has major defects.

- (d) The moment generating function of two independent random variables  $X$  and  $Y$  are:

$$M_X(t) = \frac{1}{27}(e^t + 2)^3 \quad \text{and} \quad M_Y(t) = \left(\frac{1}{3}e^t + \frac{2}{3}\right)^5$$

respectively. If  $Z = X + Y$ , then Find:

- (i)  $P(Z < 2)$
- (ii)  $P(0 \leq Z \leq 2 \mid Z \geq 1)$

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**Question 2.**

- (a) Define the terms sample space, elementary events, mutually exclusive events and exhaustive events. Write sample space in each of following cases and categorise it as discrete or continuous sample space:

- (i) Toss a coin repeatedly until head appears first time.
- (ii) A light bulb is manufactured. It is then tested for its life length by inserting it into a socket and the time elapsed (in hours) until it burns out is recorded.
- (iii) Toss a coin four times and observe the sequence of heads and tails obtained.
- (b) A man draws a ball from an urn containing 4 white and 2 red balls. If the ball is white, he does not return it to urn; if the ball is red, he does return it. He draws another ball. Let  $A$  be the event that the first ball drawn is white, and let  $B$  be the event that second ball drawn is white. Which of the following statement is true or false? Give reasons:
- (i)  $P(B) = \frac{3}{5}$
- (ii)  $P(B | A) = \frac{3}{5}$
- (iii)  $P(A | B) = \frac{9}{14}$
- (iv)  $A$  and  $B$  are independent
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**Question 3.**

- (a) A person visits randomly to a family of three children and counts the number of male children. If  $X$  denotes the number of male children and it is known that female child is twice more probable than male then write probability distribution of  $X$ . Draw a graph of probability distribution and cumulative probability distribution of  $X$ .
- (b) Let the pdf of  $X$  be given by:

$$f(x) = \begin{cases} 1/3 & \text{if } 0 < x < 1 \\ 2/3 & \text{if } 2 < x < 3 \\ 0 & \text{otherwise} \end{cases}$$

Find the distribution function of  $X$ . Calculate:

- (i)  $P\left(|X| < \frac{3}{2}\right)$
- (ii)  $P[(2x + 1) < 4]$
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**Question 4.**

- (a) Let joint pmf of two random variables  $(X, Y)$  be given by:

$$P(x, y) = \begin{cases} 1/6 & \text{if } (x, y) = (-1, -1), (-1, 1), (1, -1), (1, 1) \\ 1/12 & \text{if } (x, y) = (0, -1), (-1, 0), (0, 1), (1, 0) \\ 0 & \text{otherwise} \end{cases}$$

Find the correlation coefficient between  $X$  and  $Y$ .

- (b) Define expectation of a random variable. Show that  $E(X+Y) = E(X) + E(Y)$ . Suppose that  $X$  is a continuous random variable with the following pdf:

$$f(x) = \begin{cases} \frac{e^x}{2} & \text{if } x \leq 0 \\ \frac{e^{-x}}{2} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Find expectation of  $Y = |X|$ .

### Question 5.

- (a) Define moment generating function. Obtain moment generation function and hence mean of a random variable  $X$  having pdf:

$$f(x) = \begin{cases} \theta e^{-\theta x} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- (b) Define probability generating function. Obtain probability generating function for following distributions:

$$(i) p(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \dots, n; \quad 0 < p < 1$$

$$(ii) p(x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots, \infty; \quad \lambda > 0$$

### Question 6.

- (a) Write the statement of Chebyshev's inequality and provide its proof. Let  $X$  be a random variable with  $E(X) = 2$  and  $\text{Var}(X) = 4$ . Find lower bound of  $P(-4 \leq X \leq 8)$ .
- (b) Define normal distribution and show that its median and modes are same. In a group of individuals characteristics  $X \sim N(5, 4)$ . If we select 2 individuals at random then what is probability that the value  $X$  is greater than 5 for exactly one of them.

### Question 7.

- (a) Let  $X \sim b(n, p)$ . Show that:

$$\mu_{r+1} = pq \left( nr\mu_{r-1} + \frac{d}{dp} \mu_r \right)$$

If  $P(X = 0) = P(X = 1)$  and variance of  $X$  is  $1/4$ , then find the value of  $n$  and  $q = 1 - p$ .

- (b) Show that Poisson distribution is limiting case of binomial distribution. If  $X$  and  $Y$  are independent Poisson variate with  $P(X = 1) = P(X = 2)$  and  $P(Y = 2) = P(Y = 3)$ , then find variance of  $X - Y$ .

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